

Lesson 2 · Turning Points and Optimisation

Learning intention

Use $f'(x) = 0$ to find turning points, decide maximum or minimum, find where a function increases or decreases, and solve maximum and minimum problems.

Success looks like

- I can find a turning point and justify its nature.
- I can give an interval where f is increasing or decreasing.
- I can set up and solve an optimisation problem, with units.

Do Now

1 Differentiate $y = (x - 3)(x + 5)$.

3 Solve $3x^2 - 12 = 0$.

2 Find the gradient of $f(x) = x^3 - 5x$
at the point where $x = 2$.

4 Solve $2x^2 - x - 6 = 0$.

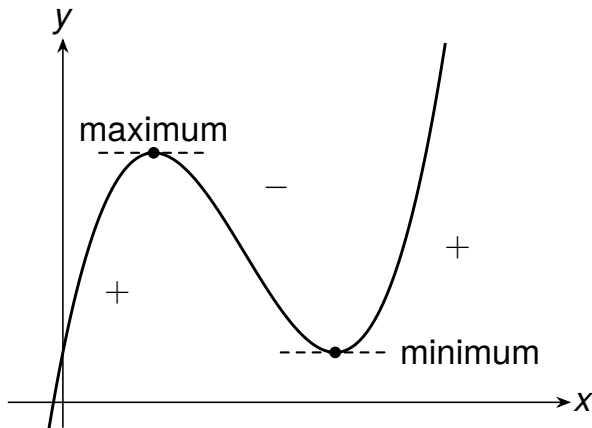
The one idea

At a turning point the tangent is flat:

$$f'(x) = 0$$

Maximum: gradient $+$ then $-$.

Minimum: gradient $-$ then $+$.



Finding turning points

Write it in this order

1. Differentiate.
 2. Set $f'(x) = 0$; solve for x .
 3. Find each y from $f(x)$.
 4. Maximum or minimum? Say why.
- 1** Find the turning points of $f(x) = x^3 - 6x^2 + 9x + 1$ and state their nature.

Maximum or minimum?

test	what you see	conclusion
gradient each side	$+$ $\rightarrow$ 0 $\rightarrow$ $-$	
	$-$ $\rightarrow$ 0 $\rightarrow$ $+$	
second derivative	$f''(x) < 0$	
	$f''(x) > 0$	

$f''(x)$: differentiate twice. Always **write the reason** in a sentence.

Practice

- 1 Find the turning point of $y = x^2 - 8x + 3$ and state whether it is a maximum or a minimum.
- 2 Find the turning points of $f(x) = 2x^3 - 3x^2 - 12x$ and justify their nature.
- 3 The curve $y = x^3 + kx$ has a turning point at $x = 2$. Find k .

Increasing and decreasing

increasing: $f'(x) > 0$ decreasing: $f'(x) < 0$

Find where $f'(x) = 0$ first; test the intervals between.

An up-parabola $f'(x)$ is **negative between its roots**.

2 For $f(x) = x^3 - 6x^2 + 9x + 1$, find where f is decreasing.

3 For which x is $y = 5 + 4x - x^2$ increasing?

Practice

- 1 For which values of x is $f(x) = x^2 - 10x + 1$ decreasing?
- 2 For which values of x is $f(x) = x^3 - 12x + 4$ decreasing?
- 3 Show that $f(x) = x^3 + 3x + 1$ is increasing for all values of x .

Optimisation

Write it in this order

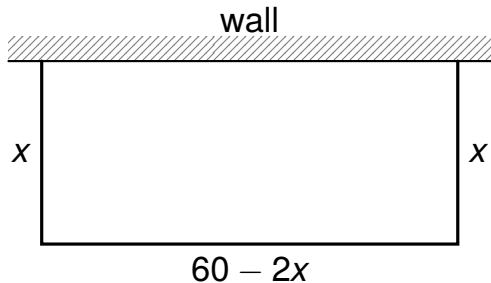
1. Formula in **one variable**.
2. Differentiate; set $= 0$; solve.
3. Justify maximum or minimum.
4. Answer the question asked, with units.

4 A ball's height is $h = 2 + 19.6t - 4.9t^2$ metres after t seconds. Find the maximum height.

Worked example · build the formula

5 A farmer has 60 m of fence to make a rectangular pen against a wall, fencing three sides. Find the maximum area.

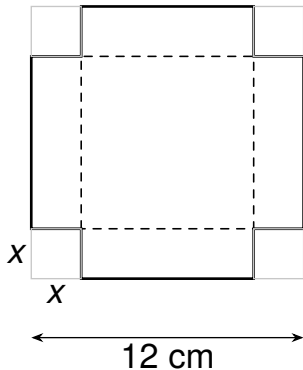
Step 1: area in terms of x only.



Practice

1 The profit, in dollars, from selling x items is $P = 120x - 2x^2 - 500$. How many items give the maximum profit, and what is that profit?

2 An open box is made from a 12 cm by 12 cm sheet by cutting a square of side x cm from each corner and folding up the sides. Find the value of x that gives the maximum volume, and the maximum volume.



Ways to lose marks

- L1 A turning point needs its y -value too.
- L2 “Maximum” needs a test and a sentence.
- L3 Increasing or decreasing is an interval.
- L4 One variable before you differentiate.
- L5 Answer the quantity asked, with units.

Summary

1. Turning point: $f'(x) = 0$; then y from $f(x)$.
2. Nature: $f'' < 0$ maximum, $f'' > 0$ minimum.
3. Increasing $f' > 0$; decreasing $f' < 0$: an interval.
4. Optimise: one variable, differentiate, $= 0$, justify, answer.

Practice sheet · 30 minutes

Work alone on the practice sheet.

- Questions 1 and 2: warm-up, about 6 minutes.
- Questions 3 to 6: exam questions, about 20 minutes.
- Question 7: Extension, if time allows.

Every maximum needs a reason; every answer needs units.

Homework

Homework sheet: 14 questions, about 60 minutes

- Questions 1 to 11: Achieved practice.
- Questions 12 and 13: Merit.
- Question 14: Extension (Excellence).

Optional extra: StudyTime Guide, “Calculating Turning Points” from page 16.